Promo Finder: finding ad-ready promos in microdrama screenplays

[Your name] · [Roll no.] · [Branch] · Problem 1 · Prototype: [GitHub link] · Raw outputs: [Drive link] · Video: [link]

1. Why Problem 1, and how I scoped it

Promos are how a microdrama gets its viewers. Kuku ships a new show every week, and today an editor has to watch every episode to find them, so this one step caps how many shows can be promoted. It's also the only problem with real inputs I could check my output against. I scoped it to the brief's literal job: **find the moments and their start/end points**, then rank them. No editing. The output is an editor's shortlist, not the final ad.

Assumptions I made where the brief was silent

- Screenplay timestamps are episode-relative and match the final cut. Clips are cut only at **CAMERA shot boundaries** and never mid-line, so every clip is something an editor can actually cut.
- "Hook in the first 3s" is judged from dialogue, action and SFX. I can't see the frames, so visual-only hooks get less credit (see failures).
- Multi-segment promos stay within one episode. I tried joining episodes, but an episode-N cliffhanger followed by episode N+1 *resolves* it, which kills the cliffhanger. Instead I built the common microdrama-ad "flash-forward": the climax beat (2–5s) first, then the setup, cutting just before the climax lands again.
- New viewers land on episode 1, so for *posting order*, promos from early episodes beat late ones of the same quality (faster payoff, fewer spoilers).

2. How it works

```
50 .txt —parse—▶ 3,727 timed events —cut—▶ 14,317 candidate clips (20–60s, shot-aligned, + flash-forwards)
        —rules—▶ hook · cliffhanger · standalone · intensity · length —select—▶ best 2 non-overlapping / episode
        —LLM (optional)—▶ cold-viewer re-judge of top 40 —rank—▶ quality × funnel weight —▶ report + CSV + editor sheet
```

Kept as code / rules	Given to the AI	Human in the loop
Parsing, cut points, never splitting a line, 20–60s bounds, de-duplication, posting order. These are exact and must never be wrong. The rules use the script's own emotion tags (`(Cold, Commanding)`) plus small genre lexicons for physical shocks, reveals and threats.	Judgement that rules can't make: "*would a stranger understand this?*". The LLM sees **only the clip's transcript**, no logline and no show context, so it reads it like someone scrolling past. It returns JSON scores and quotes the hook and cliffhanger. A guardrail checks those quotes actually appear in the clip and rejects invented text.	The editor gets `editor_review_sheet.csv` (keep/trim/reject) for 25 clips, instead of watching about 75 minutes of episodes. Clips where the rules and LLM disagree are flagged for a human check.

Swappable model. Changing the judge is one flag: `--llm gemini|openai|anthropic|none`, with `--model` to pick a model. The `openai` mode accepts any OpenAI-compatible base URL (Groq, OpenRouter, Ollama, or an API you provide). Calls are plain HTTPS with no SDKs, and responses are cached, so re-runs are free. Without a key, the rules-only mode still produces the full output.

3. Results and posting order (rules + Gemini 3.5 Flash, all 50 episodes)

#	Clip	Hook (first 3s)	Cliffhanger (cut point)	Why post it first
1	Ep18 0:56–1:29 (33s)	"The ledger!" as she runs back into a burning house	Beam crashes where she stood	Pure action, needs zero context, ends mid-disaster
2	Ep15 0:44–1:26 (42s)	"Get out of the way!" then she's shoved	"You want to fight us, veteran?"	Cuts right before the fight: a textbook cliffhanger
3	Ep5 flash-fwd (33s)	Mother's ledger flung into the mud	"Then you should have guarded it better."	Shows the cruelty first, then makes you wait for it
4	Ep1 0:39–1:21 (42s)	"Because he is marrying me."	"Father..."	Jilted at the altar: instantly clear, and it's episode 1
5	Ep13 0:24–0:56 (32s)	"I would rather face the gallows than watch you hurt yourself"	"No..."	Self-sacrifice plus a clear romantic stake

What the LLM changed. The rules and Gemini agree on the top 3 by quality. Gemini pulled 5 clips into the top 25 that the rules alone had missed (e.g. Ep32, Ep46), and moved Ep24's "Do you want the same punishment?" from #25 to #4. It dropped Ep7 from #5 to #19, saying the "opening line is too weak to stop a fast scroll", which is a call the rules couldn't make. It also found a better cut point for Ep11: the Constable rummaging through the coat, instead of the last line. 40 calls, 0 failures, 0 invented quotes. Both runs are in the raw outputs (`rules_only/` , `gemini/`).

4. How I know it's good, and where it fails

Automatic checks on every output clip: length 20–60s, no cut inside a line, something other than camera in the first 3s, minimum hook and cliffhanger scores, speaker count, scene changes, spoiler zone. All 25 top clips pass the hard checks. I also read the transcripts of the top 25 myself. **Where it picks badly** (all of these come from real runs):

- **Words, not meaning.** Ep6's cliffhanger was "dry wood catching fire in the hearth", a cosy scene that scored as danger because of "fire". Ep7's second pick ends on "a radiant, grateful smile", which is a resolution, not a cliffhanger.
- **Visual reveals are underweighted.** Ep50's real cliffhanger (Agnes silently witnessing the affair) is a camera move with no dialogue, so it ranked low.
- **Same character, different names.** The script calls Alexander "MYSTERIOUS BEARDED MAN" and Clarissa "LADY IN GOLD GOWN" or "ARISTOCRATIC WOMAN". That inflates the speaker count and would confuse a viewer-facing hook label.
- **My spoiler rule may be wrong for this show.** The title already says "He was a Duke", so Ep42's reveal ("You cannot be him. Alexander died!") is probably the *best* ad, not a spoiler. Spoiler weighting should be set per show.
- **The LLM rewards drama over clarity.** It promoted late-episode clips (Ep45, Ep46) that it rated only 5–6/10 for making sense on their own. Its "confusing references" list also counts plain character names, so that field is noisy.
- **No ground truth.** I have no list of the promos Kuku actually posted, so the weights (rules vs. LLM = 40/60) are hand-tuned, not learned.

5. Scale and cost

- Rules only: ~12s of CPU per 50-episode show, $0, deterministic. It runs on every show at upload.
- With an LLM: 40 calls of about 1.2k tokens each, roughly 50k tokens per show. That's cents per show on a flash-tier model, and it's cached, so re-runs are free. On the free tier it took about 15 minutes because of rate limits.
- Only the shortlist is sent to the LLM: rules cut 14k clips down to 40. Sending everything would cost about 350× more for little gain.
- A new genre means editing the lexicons at the top of `score.py`. A new screenplay format means a new parser; nothing else changes.

6. If a real team used it

- **Week 1:** editor keep-rate on the review sheet (target ≥ 60%) and minutes saved per show.
- **After posting:** 3-second hold rate measures the hook, completion rate measures length and pacing, and click/install rate measures the cliffhanger.
- Feed the kept/rejected labels and ad performance back in to **learn the weights** instead of hand-tuning them.

7. AI tools I used, and one place they got it wrong

[Personalise: e.g. Claude Code for the parser, scorer and LLM wrapper; ChatGPT or Gemini for ...]. **Where it got it wrong:** the first flash-forward version started the teaser on Juliana's gasp *after* the riding-crop strike (Ep24), so the strike also appeared in the setup and the "cliffhanger" just repeated the payoff. I caught it by reading the clip transcripts, then fixed it by grouping back-to-back beats and starting the teaser at the first one. A second bug: when the LLM invented a cliffhanger quote, the guardrail flagged it but the invented text still replaced the real one in the output. I fixed that too. A third: the AI hard-coded a Gemini model that Google had already retired. Now the tool takes a fallback chain of models.